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Characterization of active lead molecules from *Lissocarinus orbicularis* with potential antimicrobial resistance inhibition properties

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ABSTRACT

Background: Marine organisms are the potential contributors of novel bioactive molecules. Nevertheless, their biodiversity and the versatility of bioactive metabolites have not been fully explored. Hence, the aim of the present study was to investigate the potentials of gut associated bacteria from a marine crab for the production of novel antibacterial compound.
Methods: Aerobic gut autochthonous bacteria isolated from marine crab (*Lissocarinus orbicularis*) collected from Pazhayar coastal area in Nagapattinam district of Tamil Nadu, India were screened for antibacterial activity. Optimization for bacterial growth and antimicrobial compound production, extraction, purification and characterization were studied.
Results: In the present study, eight morphologically distinct colonies of *L. orbicularis* gut associated aerobic bacterial isolates (Iso1-Iso8) on Zobell marine agar plate were selected. Isolates were screened for antimicrobial activity against human bacterial pathogens such as *Salmonella paratyphi*, *Vibrio cholera*, *Vibrio parahaemolyticus*, *Aeromonas hydrophila* and *Listeria monocytogenes.* On the basis of screening results, isolate 5 (Iso5) was selected as the most potential strain and identified as *Paenibacillus polymyxa* using biochemical and 16S rRNA sequencing methods. The sequence data was submitted to NCBI (Gene bank Accession No: MK583465). Optimization of *P. polymyxa* for growth and antimicrobial compound production revealed incubation period (36 h), agitation (150 rpm), pH 8.0, 35 °C, 2.5% salinity, 2% glucose and 1% yeast extract as carbon and nitrogen sources respectively were the ideal conditions and mass culture was done with these parameters. Antimicrobial compound from the cell free supernatant of mass culture medium was extracted using ethanol. The lowest minimum inhibitory concentration (MIC) of 16 μg/ml was observed against of both *V. parahaemolyticus* and *V. cholerae*. GC–MS analysis of the active ethanol fraction showed the presence of different components such as dodecane (96.72%), Tridecane (1.69%), Undecane, 2,6-dimethyl- (1.69%), Tetradecane (1.12%) and Dodecane, 2,6,11-trimethyl- (1.12%).
Conclusion: The present study showed that the gut associated autochthonous bacteria of marine crabs are one of the potential sources of antibacterial compound. However, further studies are needed for the identification of the antimicrobial compound.

Introduction

Ocean covers almost 70% of the earth surface holding 80% of the earth's life forms including the vast diversity of microbes have huge potentials for finding novel natural products [1]. It is estimated that marine environments may contain a total of approximately 3.67 $\times 10^{30}$ microorganisms including the subsurface level (Whitman

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et al., 1998). The potentials of marine organisms for the discovery of new bioactive compounds are ever increasing [2,3]. However, their biodiversity and the enormous potentials of their bioactive compounds are yet to be explored. Bioactive molecules from natural sources got major attraction recently for their broad range of biological activities like antibiotic, antiviral, antioxidant, antitumor and anti-inflammatory activities [4,5]. As antibiotic resistance has become a major threat in various plants, animals and human, exploitation of the huge potential available in the microbes to produce new drugs is an urgent concern of the scientific community.

Among various microbes in the marine environment, *Bacillus* spp. seem to be omnipresent as they can quickly adapt to unfavorable conditions like high pH, temperatue, salinity and pressure by producing resistant spores [6]. Though, over 4000 antibiotics have been identified from different microorganisms so far, only 50 have been upgraded to commercial drugs for animal and plant diseases [7]. These antibiotics are highly varied in their biochemical composition, structure, mode of action, potential as well as in sources.

Recently, the gut microbiome of marine animals is considered as the potential source of bioactive compounds. In the nutrition, physiology and welfare of the animals; intestinal microflora and their metabolic activities play a vital role [8]. The microflora of crustaceans gut can metabolize a number of nutrients in to beneficial products especially when the host cannot convert them [9,10].

In the present study, gut microflora of a marine crab *Lissocarcinus orbicularis* were screened for the production of antibacterial compounds active against bacterial pathogens. In the battle of fighting against the pathogens entering through water and food, the role of gut associated microbes is also expected. Hence the present attempt on the gut microflora of the selected crab.

Materials and methods

Isolation of L. orbicularis *gut associated aerobic bacteria*

Marine crab samples *L. orbicularis* (Dana, 1852), commonly known as "Sea cucumber crab were collected from Pazhayar in Nagapattinam district of Tamil Nadu, India. After dissecting the animals, except the mid gut gland hepatopancreas, all the other gut portions (foregut, mid gut and hind gut) from each animal was removed and washed with 1.5 ml of phosphate buffer (pH-7) for 2–3 times. The gut portions were aseptically transferred into sterile centrifuge tubes and homogenized with 1 ml of phosphate buffer in a 1.5 ml micro-centrifuge tube (Tarsons, India) using a micro-pestle (Tarsons, India). After a brief centrifugation of the homogenates at 5000 rpm for 5 min, the supernatants were plated separately on Zobell marine agar for 2–3 times. The gut portions were aseptically transferred into sterile centrifuge tubes and homogenized with 1 ml of phosphate buffer in a 1.5 ml micro-centrifuge tube (Tarsons, India) using a micro-pestle (Tarsons, India). After a brief centrifugation of the homogenates at 5000 rpm for 5 min, the supernatants were plated separately on [11] Zobell marine agar (Hi Media, Mumbai, India) prepared in 50% aged-seawater for the isolation of aerobic forms of gut autochthonous bacteria. Plates were incubated at 30 ± 2 °C for 24–48 h. The density bacteria in the gut samples were expressed as Colony Forming Units (CFU) per gram of samples. CFU g^{-1} = Total number of colonies/Total volume of the sample × Volume of sample plated (0.1 ml) × dilution factor. Morphologically distinct colonies were isolated and stored at 4 °C for further antimicrobial activity screening studies.

Screening for antibacterial activity against human bacterial pathogens

Eight morphologically distinct gut bacterial isolates (designated as Iso1 to Iso8) were screened for the antimicrobial activity against 16 different human bacterial pathogens obtained from American type culture collection (ATCC) and Microbial type culture collection (MTCC), India. *Vibrio cholerae* (non – 01 non 0139 – MTCC-3906), *Escherichia coli* (ATCC-25922), *Salmonella typhi* (MTCC-3224), *Salmonella paratyphi* (MTCC-3220), *Vibrio cholerae* (01 classical – MTCC-3904), *Aeromonas hydrophila* (ATCC-7966), *Listeria monocytogenes* (MTCC-657), *Shigella boydii* (MTCC-11947), *Vibrio parahaemolyticus* (MTCC-451), *Shigella flexneri* (MTCC-1457), *Shigella dysnteriae* (ATCC-12022), *Salmonella entricatyphi* (ATCC-98), *EPEC* (Entero pathogenic *E. coli* – MTCC-443), *ETEC* (Entero toxigenic *E. coli* – MTCC-452), *V. cholerae* (EITOR- ATCC-39315) and *Klebsilla pneumonia* (ATCC-700603). Bacterial inoculum was prepared according to National committee for Clinical Laboratory Standards Institute [12] by inoculating a loopful of each test organism into individual 5 ml of nutrient broth tubes and incubated at 37 °C for 3–5 h till turbidity reaching 0.5 McFarland standards (*i.e.*) approximately 10^8 CFU/ml. Overnight bacterial cultures of isolates (Iso1-Iso8) in Zobell marine broth were incubated at 37 °C. The cell free supernatants were obtained by centrifugation at 10,000 rpm for 10 min. and used for testing antimicrobial activity on Muller Hinton agar (MHA) plates using agar-well diffusion method [13]. Strain which showed the highest zone of inhibition (diameter of zone size in centimeter) was selected as the potential strain for further study.

Identification of the potential strain

The most potential strain was identified based on standard biochemical methods described by Buchanan et al., 1974 and 16S rRNA sequencing using the bacterial primers 16SRS-F (5′-CAGGCCTAACACATGCAAGTC-3′) and 16SRS-R (5′- GGGCGGWGT GTACAAGGC-3′). PCR amplification was done using sequencing Kit (BigDye Terminator v3.1, Applied Biosystems, USA) in a thermal cycler (GeneAmp PCR System 9700, Applied Biosystems). PCR product was purified (Qiagen PCR purification kit) and sequenced (ABI 3730 DNA Analyzer, Applied Biosystems, USA). Sequence alignment was done using Geneious Pro v5.6 as described by Drummond et al. [14]. Phylogenetic tree was constructed using partial 16S rRNA gene sequences of the isolate using Neighbor joining method as described by Das and Tiwary, 2013 using MEGA 6.0 software.

Optimization of growth and antimicrobial activity

Based on the screening result, the isolate *Iso5* was selected as the most potential strain displayed antimicrobial activity against wide ranges of human bacterial pathogens tested. Various growth parameters of fermentation such as incubation time (0–48 h), agitation (0, 50, 100, 150 and 200 rpm), temperature (25 °C, 30 °C, 35 °C and 40 °C), pH (5–10 with the interval of 1), salinity (NaCl of 0.5%, 1%, 1.5%, 2%, 2.5% and 3%),1% of different carbon sources like cellulose, glucose, maltose, starch and sucrose and nitrogen sources like ammonium nitrate, ammonium sulphate, beef extract, peptone, sodium nitrate and yeast extract at 0.5% concentration were tested. The ideal carbon source glucose concentration (1.0.5%) and yeast extract as nitrogen source (0.5–2.5%) has also been tested. Growth was measured by taking the absorbance at 600 nm for every 6 h using UV spectrophotometer (Systronics, Double beam spectrophotometer 2202, India). Antimicrobial compound production was assessed based on the antimicrobial activity which was measured by the formation of zone of clearance against the bacterial pathogens using well agar diffusion assay [13]. Antimicrobial

Fig. 1. Anterior and posterior views of male (A& C) and female (B&D) crab samples (*Lissocarcinus orbicularis*, Dana. (1852)).

activity was tested against the most sensitive human pathogens *V. cholerae* and *V. parahaemolyticus* selected based on previous screening results. For mass cultivation, optimized conditions; incubation 36 h, agitation 150 rpm, pH 8.0, 35 °C, salinity 2.5%, glucose 2.0% and yeast extract 1.0% were used. Carbon and nitrogen sources were replaced in the Zobell marine broth in a 1 l conical flask with 750 ml of medium.

Extraction of antimicrobial compounds

Extraction of antimicrobial compound from the cell free supernatant (CFS) of mass culture medium was achieved by centrifugation of broth culture at 10,000 rpm for 5 min at 4 °C. The CFS was extracted with equal volume of ethanol 1:1 (v/v). After a vigorous shaking, the sample was allowed to settle and the organic phase was separated. This was repeated for 2–3 times to extract the compounds completely. The solvent phases were pooled together and concentrated by evaporation under reduced pressure using a vacuum pump (R-114, Buchi, Switzerland). Extract was tested for antimicrobial activity using agar well diffusion assay [13].

Purification of the antimicrobial compounds

The extracted antimicrobial compound was purified using silica gel (200–400 mesh) column chromatography (1.5 × 15 cm) using methanol: water (80:20) elution at the rate of 0.25 ml per min at room temperature. 16 fractions were collected and dried using rotary evaporator. The fractions were tested for antimicrobial activity.

Minimum inhibitory concentration (MIC) of the active fractions against human bacterial pathogens

The minimum inhibitory concentration (MIC) of the potential antimicrobial compound was done using agar well diffusion method described by the National committee for Clinical Laboratory Standards Institute [12]. Inoculums of the human bacterial pathogens were prepared as previously mentioned in Section "Extraction of antimicrobial compounds". After swabbing each test culture on the surface of the Muller Hinton agar plates, 6 mm wells were made on the agar plate using a sterile pipette tip then each wells were loaded with 50 μl of different concentrations (0.25, 0.5, 1, 2, 4, 8, 16, 32, 64 and 128 μg/ml) of the antimicrobial compounds. Plates were incubated at 37 ± 2 °C for 24–48 h. The formation of zone of clearance around the well indicates antibacterial activity and the MIC of the compound is the concentration at which no visible growth was observed (or) that completely inhibited the growth [15].

Characterization of antimicrobial compounds using GC–MS analysis

The antibacterial compounds present in 16 fractions were further characterized using GC–MS analysis (Clarus 500 Perkin-Elmer of auto system XL Gas Chromatograph) equipped with mass detector (Turbo mass gold-Perkin Elmer Turbomass 5.1 spectrometer) with 30 × 0.25 mm ID × 1 μm of capillary column. The initial temperature of the instrument was set to 110 °C and continuously maintained for 2 min. After end of 2 min, the oven temperature was increased 5 °C for every 1 min. and finally reached up to 280 °C and kept for 10 min. The temperature of injection port was maintained at 250 °C with helium flow of 1 ml/min. The ionized voltage of 70 eV and the sample injection at 10:1 under split mode was maintained. The mass spec range for scanning was set at 45–450 (*m/z*).

Results and discussion

Isolation of L. orbicularis *gut associated aerobic bacteria*

The common intestinal microflora of any host is important for the wellbeing of the animal. In this association both the host and the associate getting benefits symbiotically. The host getting the advan-

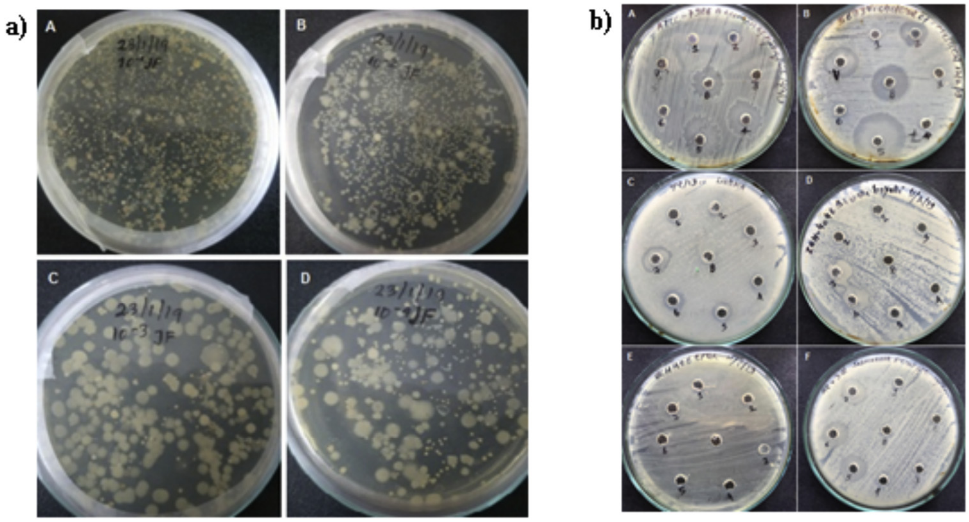

Fig. 2. a) Gut bacterial isolates of marine crab (A–D); b) screening for antimicrobial activity of *P. polymyxa* against various human bacterial pathogens (Plates A–F).

tages of assistance in the digestive metabolism, while the intestinal bacteria getting the survival advantage by nutrients, space for living and reproduction. The gut microflora and their metabolites are the prime contributor in the nourishment, functioning and welfare of animal [8]. These gut microflorae are beneficial in many ways to the host especially in the crustaceans, by metabolizing many nutrients that cannot be metabolized by the host by themselves [9,10]. The association of microbes in the guts of the several aquatic animals documented earlier, for example as endosymbionds in corals and in some invertebrates of deep sea.

In the present study, marine crabs were collected from Pazhayar landing center (Fig. 1). The aerobic forms of gut autochthonous bacteria were isolated gut homogenates plated on Zobell marine agar. The microbial density in the crab (*L. orbicularis*) gut sample was ranged from 1.6×10^2 to 1.4×10^4 CFU/g (Fig. 2a). Based on the distinct colony morphology, eight strains designated as isolates 1–8 (*Iso1-Iso8*) were selected for the screening of antimicrobial activity.

Screening of bacterial isolates for antimicrobial activity against human bacterial pathogens

On the basis of screening results, the isolate 5 (*Iso5*) displayed highest activity against wide range of pathogens tested such as *E. coli* (1.0) was 1.0 cm whereas for *S. typhi* (2.0 cm), *S. paratyphi* (1.5 cm), *V. cholerae* (01 classical) (3.0 cm), *A. hydrophila* (2.3 cm), *L. monocytogenes* (1.0 cm), *S. boydii* (1.0 cm), *V. parahaemolyticus* (3.4 cm). However, no activity was observed against other pathogens tested such as *Vibrio cholerae* (non 0139), *S. flexneri*, *S. dysnteriae*, *S. entricatyphi*, *EPEC* (Entero pathogenic *E. coli*), *ETEC* (Entero toxigenic *E. coli*), *V. cholerae* (EITOR) and *K. pneumonia* (Figs. 2b, 3 and Tables 2 and 3).

During bacterial evolution, bacteriocin production has occurred which are antagonistic to other organisms within a given competitive niche and thus provide an ecological advantage [16]. Around 4–5% of the genome in *B. subtilis* group is estimated to exclusively involving in the production of antimicrobial compounds and in it antimicrobial peptides hold the major part followed by the volatile metabolites showing many functional roles and also in metabolism [17]. Many *Bacillus* species like *B. brevis*, *B. cereus*, *B. licheniformis*, *B. polymyxa* and *B. subtilis* are reported to produce antibiotics and they are highly effective against Gram-positive bacteria [18,19]. Kaur et al., 2011 studied on the antimicrobial compound produced by *B. subtilis* which showed antimicrobial activity against *S. aureus* and *P. aeruginosa*.

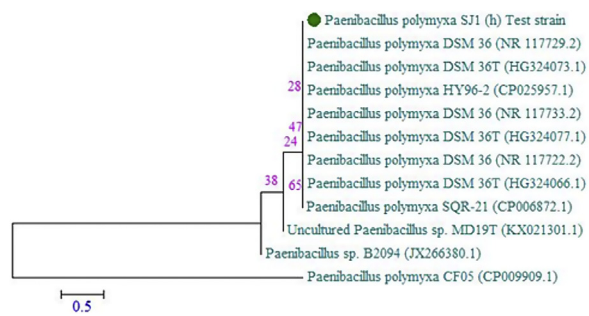

Fig. 3. Evolutionary analyses of *Panebacillus polymyxa* and phylogenetic tree construction using neighbor-joining method.

Al-Dhabi et al. [20] isolated a marine *Streptomyces* sp. strain Al-Dhabi-90 with the inhibitory potential against ESBL (extended-spectrum beta-lactamase) positive clinical pathogens. Al-Dhabi et al. [21] reported a potential antibacterial compound from a *Streptomyces* sp. strain Al-Dhabi-97 isolated from marine sediment samples sea shore of Saudi Arabia. Arasu et al. [22] reported antibacterial and antifungal activities of a marine *Streptomyces* sp. AP-123 metabolite.

Identification of the potential strain

The most potential isolate *Iso5* was identified as *Paenibacillus polymyxa* based on biochemical characterization (Table 4) and 16S rRNA gene sequencing and evolutionary relationship-based similarity analysis. The sequence data was submitted to NCBI and the Gene bank Accession number is MK583465 (Fig. 4). *P. polymyxa* is a multifunctional Gram-positive bacterium that has been reported medicinal value [23] (Yu et al., 2017).

Optimization of growth, antimicrobial activity and mass cultivation

Optimization of various physiochemical parameters for the growth and antimicrobial activity of *P. polymyxa* revealed that 36 h of incubation period, 150 rpm agitation, pH- 8.0, temperature-35 °C, salinity-2.5%, 2% glucose as carbon source and 1% yeast extract as nitrogen source were found to be the ideal conditions favored the maximum growth and antibacterial activity of the most potential strain *P. polymyxa* (Figs. 5 and 6).

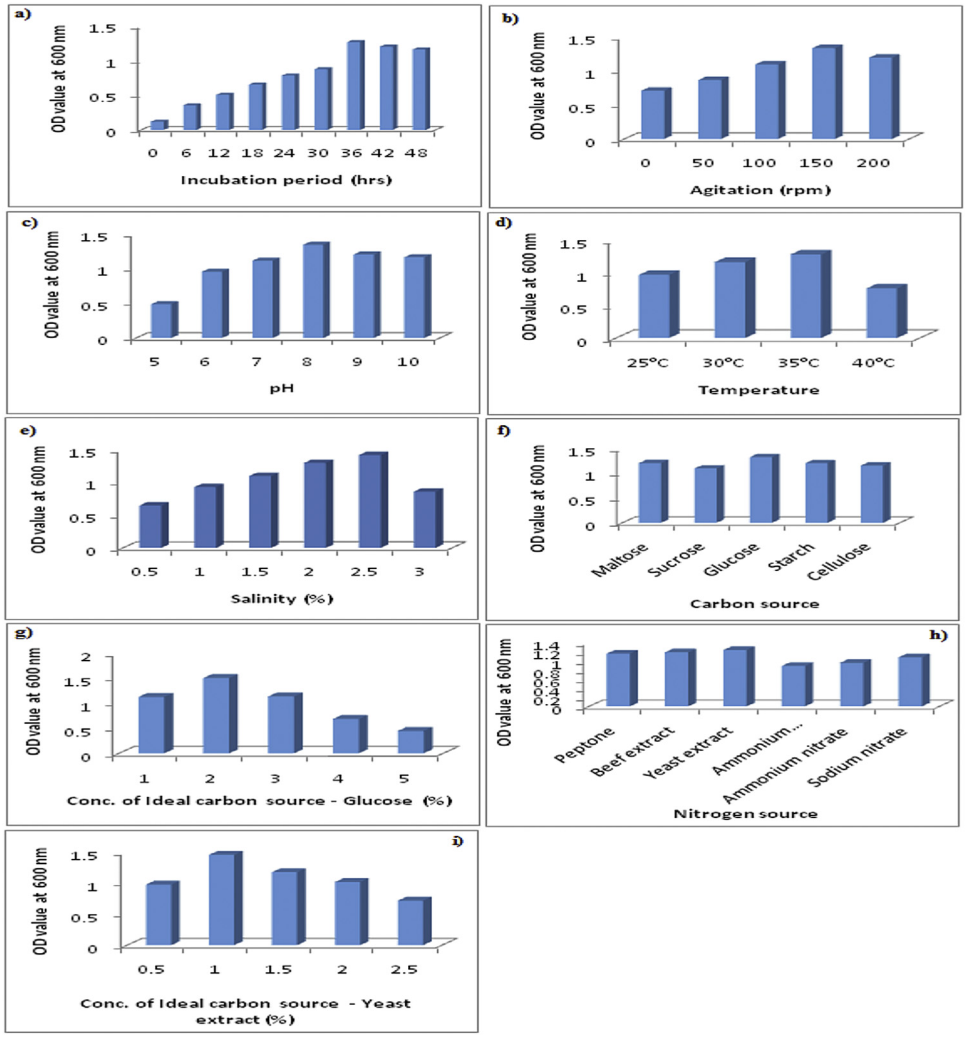

Fig. 4. Effect of various physicochemical parameters such as incubation period, agitation, pH, temperature, ideal carbon source and its concentration, ideal nitrogen source and its concentration on growth of *P. polymyxa* (a–i).

Banerjee et al., 2016 observed a positive influence of different physicochemical parameters like pH, temperature, concentration of NaCl, different carbon sources and agitation conditions on the production of antimicrobial compound in soil microorganisms. Talluri and Lanka [24] reported 72 h as the optimum for *Brevibacillus laterosporus* Strain EA62, Kaur et al. and Usta et al. [25,26] reported highest production of antimicrobial compound at 96 h in *B. subtilis*. Compared to their study, in 36 h itself highest production of antimicrobial compound was observed in the present study. Regarding effect of pH and temperature on antimicrobial compound production, Talluri and Lanka [24] and Usta et al. [26] found pH 7, 37 °C as the optimum, similarly, pH 8, 28 °C [25], pH 9 [27] as the optimum. Anjhana and Sasikala [28] studied the antagonistic *Bacillus subtilis* MS21 from Thengapattanam estuary against various plant fungal pathogens found 1% salinity as the ideal concentration for antimicrobial compound production. Tabbene et al. [29] optimized the medium composition for the production of antimicrobial activity by *Bacillus subtilis* B38 found 1.5% lactose as the ideal carbon. Kaur et al. [25] found highest zone of clearance of 3.96 cm by *B. subtilis* grown in peptone as the nitrogen source. Ju et al. [30] observed 3% glucose as the ideal carbon source. Usta et al. [26] found glucose as the ideal carbon source and yeast extract as the ideal nitrogen source for the antimicrobial compound production by *B. laterosporus* Strain EA62.

Al-Ansari et al. [31] observed that pH, temperature and NaCl concentration play a significant role in the growth and antibiotic production in a marine environmental *Streptomyces* sp. AS11 isolate. They found pH 7.3; 29 °C and 34% seawater enhance the growth while pH 7.5, 38 °C, glucose and starch as carbon source and sodium nitrate as the nitrogen source gave maximum production of antibiotics in the medium. Sebak et al. [32] optimized the production of antimicrobial compounds from a *Streptomyces* sp. MS. 10 strain isolated from Egyptian soil found broad-spectrum of antimicrobial activity against bacterial pathogens including methicillin-resistant *S. aureus* (MRSA).

With these optimized growth parameters mass scale production in was done in a shake flask culture. Talluri and Lanka [24] observed 1.2-fold increased antimicrobial compound production with the optimum medium when compared to the basal medium.

Extraction of antimicrobial compounds

The antimicrobial compound in the cell free supernatant of mass scale culture medium of *P. polymyxa* was extracted using ethanol as the solvent. The results showed that between the aqueous and organic phase; the antimicrobial compounds were present in the organic phase. Hence it was used for further study.

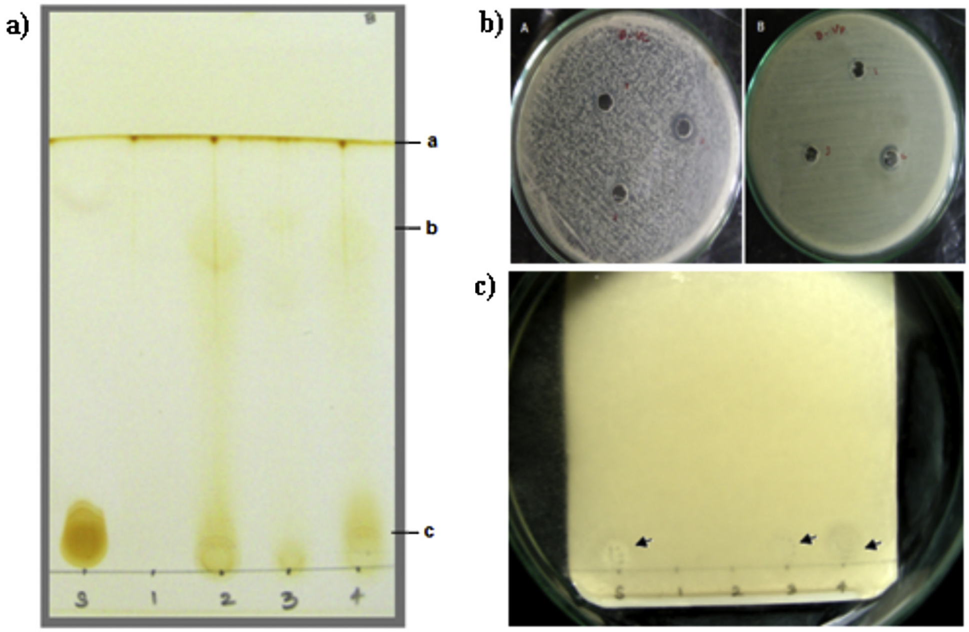

Fig. 5. a) Separation of antimicrobial compounds extracted from the cell free extract of *P. polymyxa* using thin layer chromatography (TLC) where 'a – Solvent front, b and c – separated compounds, S – Standard, 1-4: different solvent extracts *viz.*, chloroform, ethyl acetate, ethanol and methanol respectively'. **b)** Antimicrobial activity of TLC purified antimicrobial compounds spot against *V. cholerae* and *V. parahaemolyticus* (Plates A and B respectively). **c)** Bioautography of antimicrobial compounds from the cell free extract of *P. polymyxa* on TLC plate against *V. parahaemolyticus*.

Table 1
GCMS analysis of antimicrobial compound from *P. polymyxa* - Overall library search results table.

Compound name	RT	Molecular formula
[1,1′-Bicyclopropyl]-2-octanoic acid,2′-hexyl-, methylester	5.53	$C_{21}H_{38}O_2$
3-Cyclopentene-1-ethanol,2,2,4-trimethyl-	5.53	$C_{10}H_{18}O$
4,8-Decadienal, 5,9-dimethyl-	5.53	$C_{12}H_{20}O$
3Dodecene, (E)	5.91	$C_{12}H_{24}$
1Dodecene	5.91	$C_{12}H_{24}$
Cyclododecane	5.91	$C_{12}H_{24}$
Isobornylthiocyanoacetate	6.05	$C_{13}H_{19}NO_2S$
1,2-Cis-1,5-trans-2,5-dihydroxy-4-methyl-1-(1-hydroxy-1-isopropyl)cyclohex-3-ene	6.05	$C_{10}H_{18}O_3$
1,2-Trans-1,5-trans-2,5-dihydroxy-4-methyl-1-(1-hydroxy-1-isopropyl)cyclohex-3-ene	6.05	$C_{10}H_{18}O_3$
1,2-Cis-1,5-trans-2,5-dihydroxy-4-methyl-1-(1-hydroxy-1-isopropyl)cyclohex-3-ene	6.15	$C_{10}H_{18}O_3$
3-Cyclopenten-1-one,2,2,5,5-tetramethyl-	6.15	$C_9H_{14}O$
Bicyclo[3.1.0]hexane,1,5-dimethyl-	6.15	C_8H_{14}
3-Trifluoroacetoxypentadecane	6.32	$C_{17}H_{31}F_3O_2$
2-Trifluoroacetoxytridecane	6.32	$C_{15}H_{27}F_3O_2$
3-Trifluoroacetoxydodecane	6.32	$C_{14}H_{25}F_3O_2$
Dodecane	6.41	$C_{12}H_{26}$
Tridecane	6.41	$C_{13}H_{28}$
Undecane,2,6-dimethyl-	6.41	$C_{13}H_{28}$
Tetradecane	8.30	$C_{14}H_{30}$
Dodecane,2,6,11-trimethyl-	8.30	$C_{15}H_{32}$
Dodecane	8.30	$C_{12}H_{26}$
Phenol, 2,6-bis(1,1-dimethylethyl)-	9.33	$C_{14}H_{22}O$
Phenol, 2,4-bis(1,1-dimethylethyl)-	9.33	$C_{14}H_{22}O$
Phenol, 3,5-bis(1,1-dimethylethyl)-	9.33	$C_{14}H_{22}O$
Dodecane,2,6,11-trimethyl-	9.95	$C_{15}H_{32}$
Pentadecane	9.95	$C_{15}H_{32}$
Decane,2,4,6-trimethyl-	9.95	$C_{13}H_{28}$
2,2-Bis[4-[[4-chloro-6-(3-ethynylphenoxy)-1,3,5-triazin-2-yl]oxy]phenyl]propane	30.38	$C_{37}H_{24}C_{l2}N_6O_4$
Pregn-5-en-20-one,3,16,17,21-tetrakis[(trimethylsilyl)oxy]-, *O*-(phenylmethyl)oxime, (3á,16à)-	30.38	$C_{40}H_{71}NO_5Si_4$
2,2-Bis[4-[(4,6-dichloro-1,3,5-triazin-2-yl)oxy]ph enyl]-1,1,1,3,3,3-hexafluoropropane	30.38	$C_{21}H_8Cl_4F_6N_6O_2$

Purification of the antimicrobial compounds

Compounds which were having inhibitory activity against the most sensitive pathogens present in the extract were purified using column chromatography. Among the 16 fractions collected active compounds were present the fractions 3 to fraction 8. The active fractions were pooled together and concentrated using rotary evaporation and the purity of the compound was tested using TLC analysis.

Determination of minimum inhibitory concentration (MIC) of the potential strain against human bacterial pathogens

The antimicrobial compound produced from *P. polymyxa* was quantitatively assessed for determining the MIC against human

Table 2
Qualitative determination of antimicrobial activity of marine bacterial isolates against human bacterial pathogens.

Pathogens tested	Antimicrobial activity of the isolates (Iso1 – Iso8)							
	Iso1	Iso2	Iso3	Iso4	Iso5	Iso6	Iso7	Iso8
Shigella boydii	–	–	–	–	+	+	+	–
Vibrio cholerae classical	–	+	–	+	+	–	–	+
Salmonella entrica typhi	–	–	–	–	+	+	+	–
Vibrio parahaemolyticus	–	–	–	–	+	+	–	–
EPEC	+	–	–	–	+	+	+	–
Aeromonas hydrophyla	–	+	+	+	+	–	–	+
Vibrio cholerae (EITOR)	–	–	–	–	–	–	–	–
Klebsilla pneumoniae	–	–	–	–	–	–	–	–
ETEC	–	–	–	–	–	–	+	–
Listeria monocytogenes	–	–	–	–	+	+	+	–
Shigella dysnteriae	+	–	–	–	–	+	+	–
E. coli	–	–	–	–	–	–	–	–
Vibrio cholerae (non – 01 non 0139)	–	–	–	–	–	–	–	–
Salmonella typhi	–	–	–	–	–	–	–	–
Salmonella paratyphi	+	–	+	–	+	+	+	–
Shigella flexneri	–	–	–	–	–	–	–	–

+ Positive, – negative reaction.

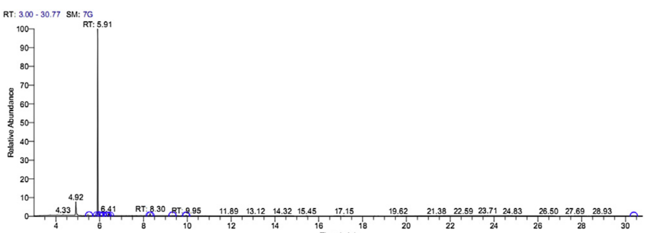

Fig. 6. GCMS analysis of antimicrobial compound from *P. polymyxa*.

Table 3
Antimicrobial activity of *P. polymyxa* against human bacterial pathogens.

Pathogens tested	Zone of clearance in diameter (cm)
Vibrio cholerae (non – 01 non 0139)	–
Escherichia coli	1.0
Salmonella typhi	2.0
Salmonella paratyphi	1.5
Vibrio cholerae (01 classical)	3.0
Aeromonas hydrophila	2.3
Listeria monocytogenes	1.0
Shigella boydii	1.0
Vibrio parahaemolyticus	3.4
Shigella flexneri	–
Shigella dysnteriae	–
Salmonella entrica typhi	–
EPEC	–
Vibrio cholerae (EITOR)	–
Klebsilla pneumoniae	–
ETEC	–

bacterial pathogens. When testing different concentrations ranged from 0.25 to 128 μg/ml, the lowest MIC 16 μg/ml was observed against *V. cholerae* and *V. parahaemolyticus* and highest the MIC was observed against *E. coli* (64 μg/ml) (Table 5). Usta et al. [26] observed MIC of >256 μg/ml against the five different bacterial pathogens tested by the antimicrobial compound produced by *B. laterosporus* Strain EA62. Wu et al. [27] observed the MIC of antimicrobial compound produced by *B. subtilis* BY08 against *B. cereus* and *L. monocytogenes* were 64 μg/ml and 128 μg/ml respectively. Al-Dhabi et al. [20] found MIC values ranging from 62.5 to 500 μg/ml for a *Streptomyces* extract against some Gram negative bacteria. Compared to the above studies the MIC values observed in the present study seemed to be lower.

Table 4
Biochemical identification of *Paenibacillus polymyxa*.

Test	Results
Gram's staining	+ve
Morphology	Rod
Motility	+
Catalase	–
Indole	–
Methyl red	–
Voges Proskauer	+
Citrate utilization	+
Urease	–
TSI	Alkaline slant
Starch hydrolysis	+
Gelatin hydrolysis	+
Casein hydrolysis	+
Glucose	+
Arabinose	+
Xylose	+
Lactose	–
Sucrose	+
Maltose	+
Manitol	+
Oxidase	+

Table 5
Minimum inhibitory concentration (MIC) of *P. polymyxa* antimicrobial compound against human bacterial pathogens.

Pathogens tested	MIC (μg/ml)
Vibrio cholerae (non – 01 non 0139)	–
Escherichia coli	32
Salmonella typhi	32
Salmonella paratyphi	32
Vibrio cholerae (01 classical)	16
Aeromonas hydrophila	32
Listeria monocytogenes	32
Shigella boydii	32
Vibrio parahaemolyticus	16
Shigella flexneri	–
Shigella dysnteriae	–
Salmonella entrica typhi	–
EPEC	–
Vibrio cholerae (EITOR)	–
Klebsilla pneumoniae	–
ETEC	64

Characterization of antimicrobial compounds using GC–MS analysis

The results revealed that the antimicrobial agents showed different peaks and the details of the compounds Table 1. GC–MS analysis showed the presence of different components such as dodecane (96.72%), tridecane (1.69%), undecane, 2,6-dimethyl-(1.69%), tetradecane (1.12%) and dodecane, 2,6,11-trimethyl-(1.12%) apart from that essential oils, phenol, triazine, isobornyl thiocyanoaceatate were important compounds found in GC analysis. Verma et al., 2015 characterized the antimicrobial compound produced by *Pseudomonas* sp. using GC–MS analysis opined that the presence of the compound naphth [2,3-B] azet-2 (1*H*)-one, 1-phenyl-, might be the reason for antibacterial activity. All these compounds or their derivatives were already reported to have antimicrobial activity. For example, Badaghu and Gavaskar [33] reported the antimicrobial potentials of spiropyrroidines derived from cyclopentanone.

Conclusion

The present study on the production of antimicrobial compounds from gut associated microbial sources from the marine crab *L. orbicularis* showed that these microbes especially the autochthonous forms could be the fair alternative for the search of newer antimicrobial compounds. The gut associated isolate *P. polymyxa* in the present work had displayed wider antimicrobial activity against a wide range of bacterial pathogens tested.

Author contributions

Conception, design collection of data and analysis were done by S. H., Data interpretation was done by, M. P., S.S., M.S. A., S. S., N. J and S. D., K.M. and A. R. drafted the article. Over all supervision of the work carried out by S. J.

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Competing interests

Authors are declare no conflicts of interest.

Ethical approval

This article does not contain any studies with human participants or animal performed by any of the authors.

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